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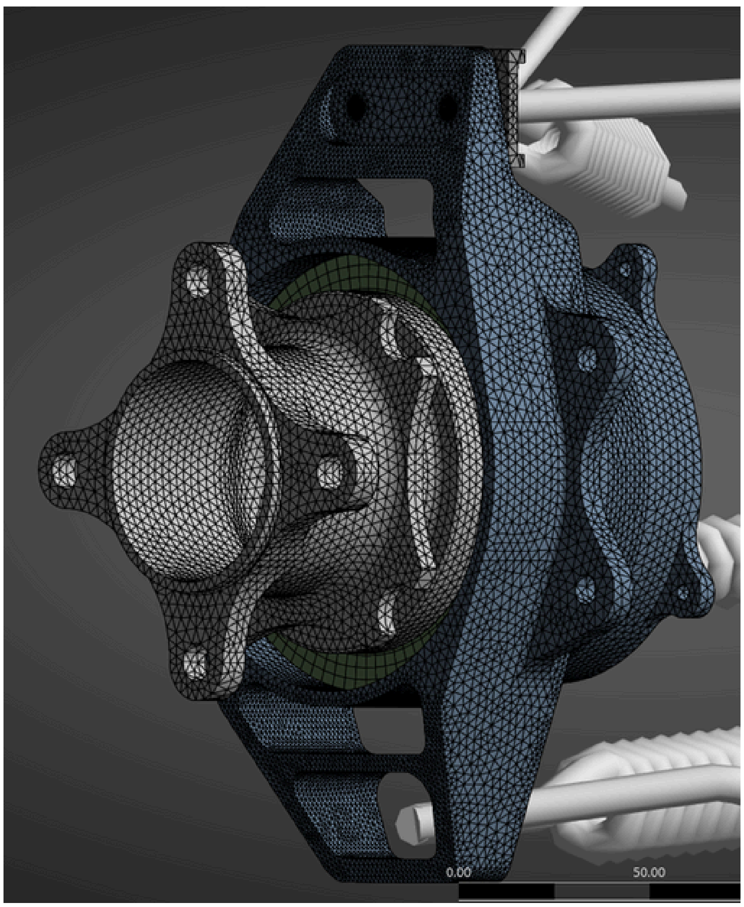
Suspension Upright

Situation: In the past, there existed a lack of stress analysis within the drivetrain assembly and knowledge of FEA software utilization.

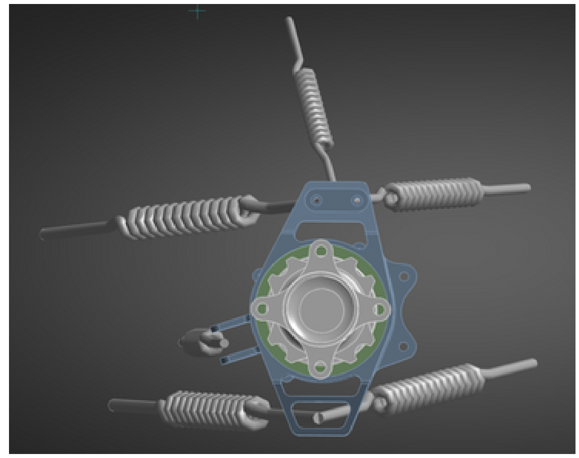
Task: Design and conduct Ansys FEA on rear and front upright designs. Establish FEA knowledge foundation for the team.



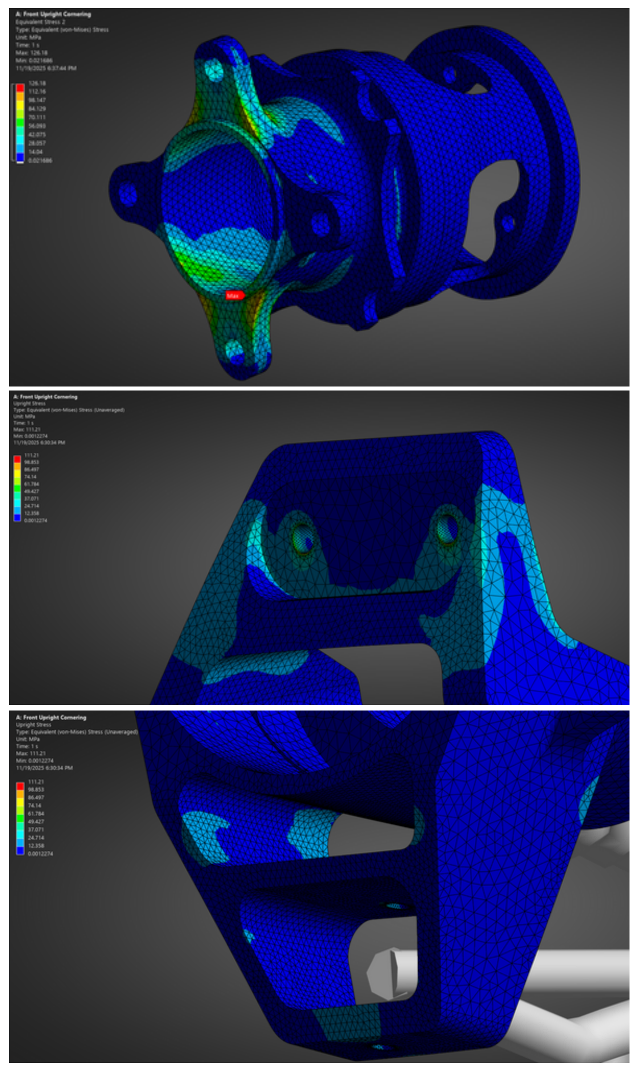
Mesh size with local sizing features at hole edges and faces experiencing the most stress.

Action: Created a full assembly setup implementing springs to represent suspension and steering arms along with the hub and bearings to simulate accurate load transfer. Performed corner and braking load cases and verified results through a mesh convergence study.

Result: Robust fea setup that produced accurate stress results on upright. Established FEA standards for future use in geometry optimization.



Assembly setup includes A arms, push rod, tie rod, bearings, hub, upright, camber piece

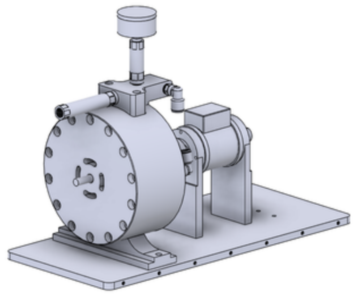


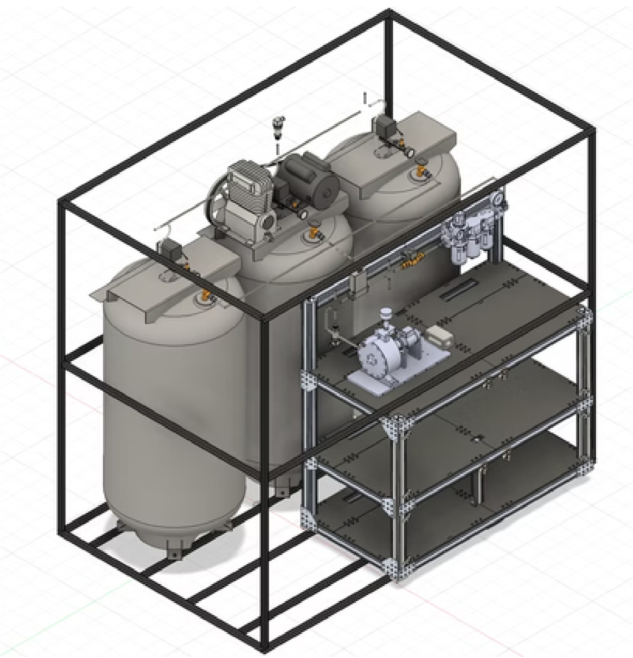
Stress appears in expected regions of upper and lower arms of upright and hub tabs

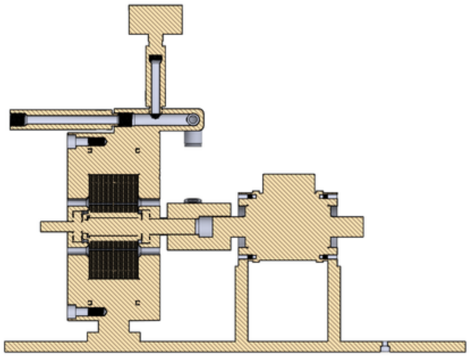
Air Battery

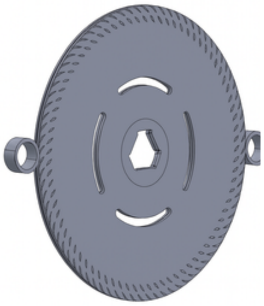
Situation: A boundary layer turbine system that utilizes an air compressor to generate power was previously validated on a smaller scale. For in field testing, the system required a robust stucture for transportation and operational use.

Task: Create a strong yet cost efficient structure to house the entire system. The structure must also be capable of withstanding vibrational cyclic loading of the turbine and overseas transportation.

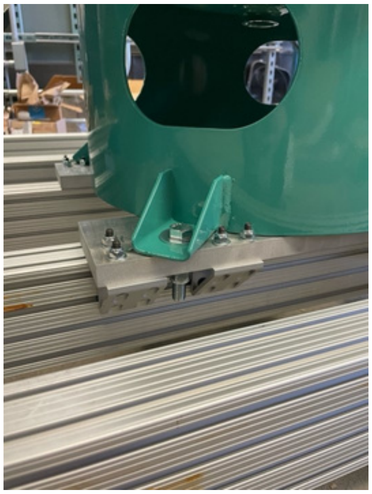


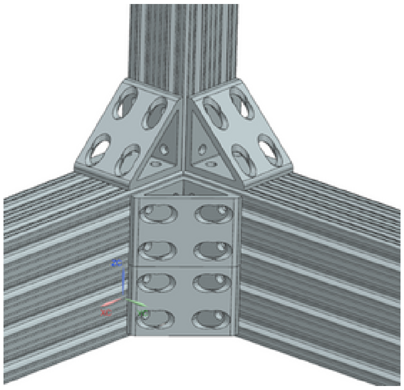






Action: Designed the structural frame (NX), sourced materials, and assembled a modular 80/20 beam structure optimized for strength and ease of assembly. Selected booster tanks and corresponding fasteners, and machined custom mounting plates to secure the tanks to the 80/20 frame.

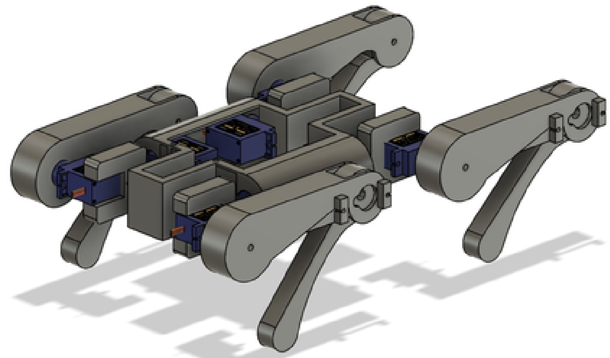




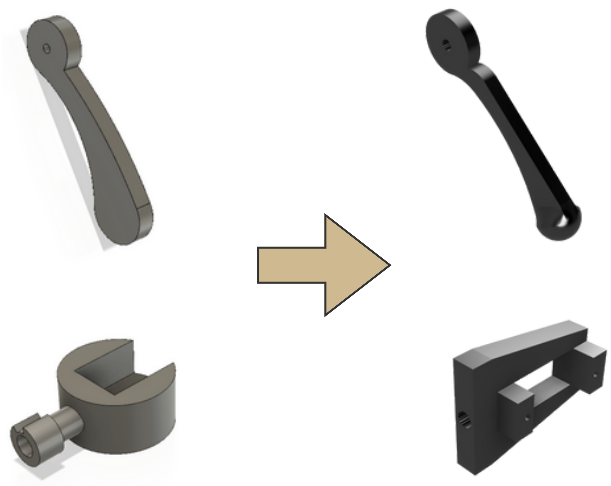
Result: Rigid support structure capable of supporting 1600lbs that features a table for turbine mounting. Elevated on 8in feet for easy forklift access and utilizes nylock fasteners to combat vibrational loosening.

Dogcopter

Situation: The autonomous robotics club created an initial concept of a quadruped robot capable of transforming into a drone.

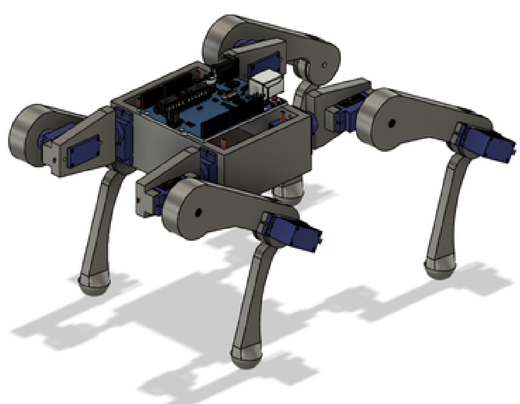


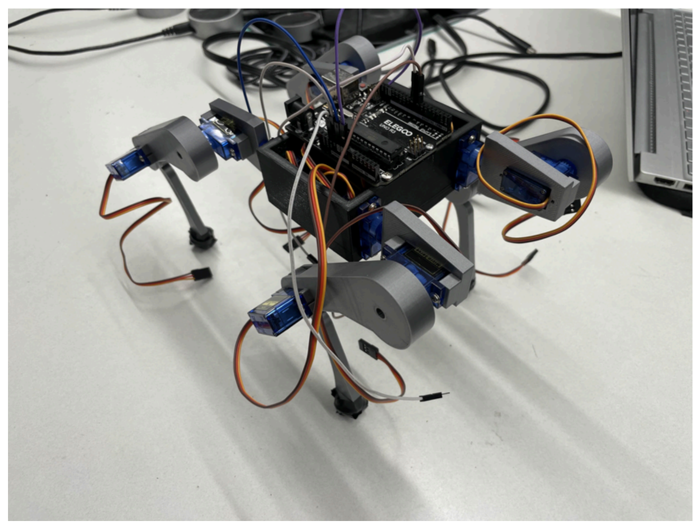
Task: Improve and fabricate the inital functional prototype to flesh out preliminary challenges before progressing to a full scale model.



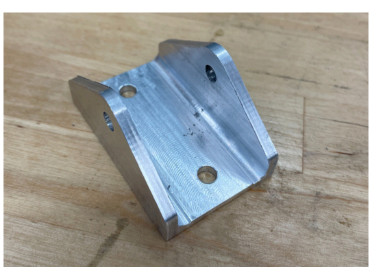
Action: Designed and created a 3d printed prototype, specifically redesigning the lower leg for greater surface area contact and hip joint for better fixture to servo.

Result: Quadruped capable of standing, kneeling, and laying down to transform into drone mode.

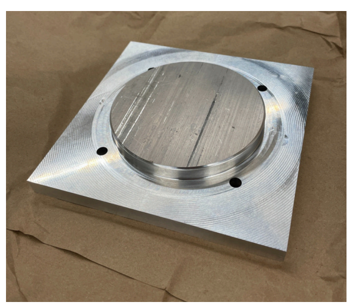


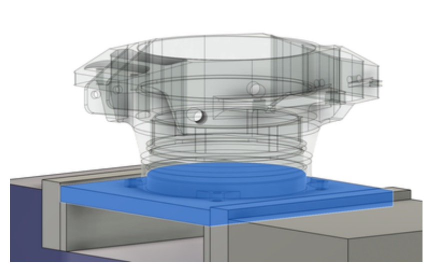


Manufacturing Projects



Upright bracket:
3 operations on 3 axis CNC





Fixture plate for Upright:
2 Ops on 3 axis CNC
Cylindrical boss feature adheres to H8 Misumi standards to maintain secure fit on the upright for final Op.